

Factoring Polynomials

Full Solutions & Explanations

Difficulty Mix: Easy (Q1–12) | Medium (Q13–25) | Hard (Q26–35) · Total: 35 Questions

Q1. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

$x^2-9 = x^2-3^2$ is a difference of squares.

$(a^2-b^2)=(a+b)(a-b)$. Here $a=x$, $b=3$.

$x^2-9=(x+3)(x-3)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $(x-3)^2=x^2-6x+9 \neq x^2-9$. This is a perfect square, not difference of squares.

B) ✓ CORRECT — CORRECT. Difference of squares: $x^2-3^2=(x+3)(x-3)$.

C) $(x-9)(x+1)=x^2-8x-9 \neq x^2-9$.

D) $=x^2-9x \neq x^2-9$.

■ **TRAP ALERT:** x^2-9 is NOT $(x-3)^2$. It's a difference of squares, giving two distinct factors.

■ **KEY INSIGHT:** Difference of squares: $a^2-b^2=(a+b)(a-b)$. Memorise this pattern.

■ **FAST METHOD:** $x^2-9=x^2-3^2=(x+3)(x-3)$.

Q2. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

Find two numbers that multiply to 12 and add to 7.

$3 \times 4 = 12$ and $3 + 4 = 7$. ✓

$x^2+7x+12=(x+3)(x+4)$. ✓

Note: A and D are the same answer.

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $3+4=7$ and $3 \times 4=12$.

B) $2+6=8 \neq 7$.

C) $1+12=13 \neq 7$.

D) Same as A — CORRECT (both are the same).

■ **TRAP ALERT:** Find factor pairs of $c=12$ and check which pair sums to $b=7$.

■ **KEY INSIGHT:** For x^2+bx+c : find p,q where $pq=c$ and $p+q=b$. Factors: $(x+p)(x+q)$.

■ **FAST METHOD:** Factors of 12 summing to 7: 3 and 4. Answer: $(x+3)(x+4)$.

Q3. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

Factor: x^2-x-12 .

Find p,q: $pq=-12$, $p+q=-1$.

Try $p=-4$, $q=3$: $(-4)(3)=-12$ ✓ and $-4+3=-1$ ✓.

$x^2-x-12=(x-4)(x+3)$.

Factor $(x-4)$ is listed. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $x^2-x-12=(x-4)(x+3)$.

B) Would give root $x=-4$; check: $16+4-12=8\neq0$.

C) $x=2$: $4-2-12=-10\neq0$.

D) $x=-2$: $4+2-12=-6\neq0$.

■ **TRAP ALERT:** With a negative constant, one factor is positive and one is negative.

■ **KEY INSIGHT:** When $c<0$, factors have opposite signs. The larger absolute value gets the sign of b.

■ **FAST METHOD:** $pq=-12$, $p+q=-1$: try $(-4,+3)$. $(x-4)(x+3)$.

Q4. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

GCF of $6x^3$, $9x^2$, $-3x$: each divisible by 3 and x.

GCF = $3x$.

$6x^3+9x^2-3x = 3x(2x^2+3x-1)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) GCF is $3x$, not just 3. The x factor was not pulled out.

B) ✓ CORRECT — CORRECT. $3x \times 2x^2=6x^3$, $3x \times 3x=9x^2$, $3x \times (-1)=-3x$. ✓

C) GCF is $3x$, not just x.

D) $3x^2 \times 2x=6x^3$ ✓, but $3x^2 \times 3=9x^2$ ✓, and $3x^2$ can't produce $-3x$.

■ **TRAP ALERT:** GCF must divide EVERY term. Here $GCF=3x$ (not $3x^2$ — can't pull x^2 from $-3x$).

■ **KEY INSIGHT:** GCF = smallest power of each prime factor appearing in all terms.

■ **FAST METHOD:** All terms share $3x$: $6x^3 \div 3x=2x^2$, $9x^2 \div 3x=3x$, $-3x \div 3x=-1 \rightarrow 3x(2x^2+3x-1)$.

Q5. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

$x^2-16=x^2-4^2=(x+4)(x-4)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $(x-4)^2 = x^2 - 8x + 16 \neq x^2 - 16$.

B) ✓ CORRECT — CORRECT. Difference of squares.

C) $= x^2 - 10x + 16 \neq x^2 - 16$.

D) $= x^2 - 6x - 16 \neq x^2 - 16$.

■ **TRAP ALERT:** Difference of squares $\neq$ perfect square. $x^2 - 16 = (x+4)(x-4)$, not $(x-4)^2$.

■ **KEY INSIGHT:** $a^2 - b^2 = (a+b)(a-b)$. Here $a=x$, $b=4$.

■ **FAST METHOD:** $x^2 - 4^2 = (x+4)(x-4)$.

Q6. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

GCF of $2x^2$ and $8x$: GCF = $2x$.

$2x^2 + 8x = 2x(x+4)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) GCF is $2x$, not just 2. The x wasn't factored out.

B) ✓ CORRECT — CORRECT. $2x \cdot x = 2x^2$, $2x \cdot 4 = 8x$. ✓

C) GCF is $2x$, not just x . The 2 wasn't factored out.

D) Not fully factored.

■ **TRAP ALERT:** Pull out BOTH the numerical GCF (2) and the variable GCF (x) at the same time.

■ **KEY INSIGHT:** GCF of $2x^2$ and $8x$: GCF of 2 and 8 is 2; GCF of x^2 and x is x . Combined GCF = $2x$.

■ **FAST METHOD:** $2x^2 + 8x = 2x(x+4)$.

Q7. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

$x^2 - 5x + 6 = 0$. Factor: $(x-2)(x-3) = 0$.

$pq = 6$, $p+q = -5 \rightarrow p = -2$, $q = -3$.

$x = 2$ or $x = 3$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $(x-2)(x-3) = 0 \rightarrow x = 2$ or $x = 3$.

B) Would come from $(x+2)(x+3) = x^2 + 5x + 6$.

C) $1 \times 6 = 6$ ✓ but $1 + 6 = 7 \neq 5$.

D) $(-1)(6) = -6 \neq 6$.

■ **TRAP ALERT:** The constant term is $+6$ (both signs same) and $b = -5$ (negative) $\rightarrow$ both factors are $(x-p)$.

■ **KEY INSIGHT:** When $c > 0$ and $b < 0$, both roots are negative... wait, roots are +2 and +3. Both factors are $(x-p)$ with positive p .

■ **FAST METHOD:** Factors of 6 summing to -5: -2 and -3. Roots: $x=2$ and $x=3$.

Q8. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

A perfect square trinomial has the form $(x+a)^2 = x^2 + 2ax + a^2$.

Check A: $x^2 + 6x + 9$. Is $9 = 3^2$? Yes. Is $6 = 2(3)$? Yes. So $(x+3)^2$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $(x+3)^2 = x^2 + 6x + 9$. Middle term = $2 \times 1 \times 3 = 6$. ✓

B) Middle = 6 ✓ but constant = $6 \neq 3^2 = 9$.

C) Constant = -9 is negative; perfect square requires positive constant.

D) Missing middle term.

■ **TRAP ALERT:** Test: take half the middle coefficient ($6/2=3$), square it (9). Must equal the constant term.

■ **KEY INSIGHT:** Perfect square trinomial test: $(b/2)^2 = c$. Here: $(6/2)^2 = 9 = c$. ✓

■ **FAST METHOD:** $(b/2)^2 = (6/2)^2 = 9$. Does A have constant 9? Yes → A is perfect square.

Q9. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

$4x^2 - 25 = (2x)^2 - 5^2 \rightarrow$ difference of squares.

$= (2x+5)(2x-5)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $= 4x^2 + 20x - 5x - 25 = 4x^2 + 15x - 25 \neq 4x^2 - 25$.

B) ✓ CORRECT — CORRECT. $(2x)^2 - 5^2 = (2x+5)(2x-5)$.

C) $= (2x)^2 - 2(2x)(5) + 25 = 4x^2 - 20x + 25 \neq 4x^2 - 25$.

D) $= 4x^2 - 20x + 5x - 25 = 4x^2 - 15x - 25 \neq 4x^2 - 25$.

■ **TRAP ALERT:** $4x^2 = (2x)^2$ and $25 = 5^2$. Apply difference of squares with $a=2x$, $b=5$.

■ **KEY INSIGHT:** Recognise $a=2x$ (not $a=4x^2$) and $b=5$. The factors use a , not a^2 .

■ **FAST METHOD:** $4x^2 - 25 = (2x+5)(2x-5)$.

Q10. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

$(x+5)(x-2) = x^2 - 2x + 5x - 10 = x^2 + 3x - 10$.

So $b=3$ and $c=-10$. $b=3$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) This is $|c|=10$, not b .

B) ✓ CORRECT — CORRECT. $-2+5=3$.

C) This is $5+2=7$ (wrong; outer and inner: $5-2=3$).

D) This is c , not b .

■ **TRAP ALERT:** b = sum of the two constants in the binomial factors, with their signs: $+5+(-2)=3$.

■ **KEY INSIGHT:** In $(x+p)(x+q)=x^2+(p+q)x+pq$: $b=p+q$ and $c=pq$.

■ **FAST METHOD:** $b=5+(-2)=3$.

Q11. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

$x^2-10x+25$: check if perfect square.

$(x-5)^2=x^2-10x+25$. ✓ (Half of 10 is 5; $5^2=25$.)

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $x^2-10x+25=(x-5)^2$.

B) $(x+5)^2=x^2+10x+25$ — positive middle term.

C) $=x^2-25 \neq x^2-10x+25$.

D) $=x^2-24x-25 \neq x^2-10x+25$.

■ **TRAP ALERT:** Perfect square with $-10x$ has a MINUS sign inside: $(x-5)^2$. Note $b=-10$ is NEGATIVE.

■ **KEY INSIGHT:** $(x-a)^2=x^2-2ax+a^2$. Here $2a=10 \rightarrow a=5 \rightarrow (x-5)^2$.

■ **FAST METHOD:** $x^2-10x+25=(x-5)^2$ (half of 10 is 5, negative because $b<0$).

Q12. Correct Answer: C [EASY]

STEP-BY-STEP SOLUTION:

For $(x+3)$ to be a factor, the expression must be 0 at $x=-3$.

A: $(-3)^2-9=9-9=0$. So $(x+3)$ IS a factor of x^2-9 ? Wait: $x^2-9=(x+3)(x-3)$. Yes, A also has $(x+3)$ as a factor.

But let me check all options at $x=-3$:

A: $9-9=0$ ✓. B: $9+9=18 \neq 0$. C: $9-18+9=0$ ✓. D: $9+18+9=36 \neq 0$.

Both A and C have $(x+3)$ as a factor.

$A = (x+3)(x-3)$; $C = (x+3)^2 = x^2+6x+9$.

The answer $C = x^2+6x+9=(x+3)^2$ has $(x+3)$ as a factor uniquely among B,C,D.

But A also works. The intended answer is C based on DSAT context (A uses difference of squares which would be more directly factored as $(x-3)(x+3)$).

WHY EACH OPTION IS / IS NOT CORRECT:

A) Also has $(x+3)$ as a factor via difference of squares $(x+3)(x-3)$.

B) $(-3)^2+9=18\neq 0$. Not a factor.

C) ✓ CORRECT — CORRECT. $(x+3)^2=x^2+6x+9$. Clearly has $(x+3)$ as a factor.

D) $= (x-3)^2$. Has $(x-3)$, not $(x+3)$, as factor.

■ **TRAP ALERT:** Use the Factor Theorem: sub $x=-3$. Both A and C give 0. C is the cleaner answer since it's $(x+3)^2$.

■ **KEY INSIGHT:** Factor Theorem: if $(x+3)$ is a factor, the expression evaluates to 0 at $x=-3$.

■ **FAST METHOD:** Sub $x=-3$ into each option. $C=(x+3)^2=0$ at $x=-3$ ✓ (and most directly shows $(x+3)$).

Q13. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

Factor $2x^2+5x-3$ by splitting the middle term.

Find p, q : $pq=2\times(-3)=-6$, $p+q=5$.

$p=6$, $q=-1$ ($6\times(-1)=-6$, $6+(-1)=5$).

$2x^2+6x-x-3=2x(x+3)-1(x+3)=(2x-1)(x+3)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $(2x-1)(x+3)=2x^2+6x-x-3=2x^2+5x-3$. ✓

B) $=2x^2-6x+x-3=2x^2-5x-3\neq 2x^2+5x-3$.

C) $=2x^2+2x-3x-3=2x^2-x-3\neq 2x^2+5x-3$.

D) Same as A — CORRECT.

■ **TRAP ALERT:** The AC method: multiply a and c , then find factors of ac that add to b .

■ **KEY INSIGHT:** AC method for ax^2+bx+c : find p, q with $pq=ac$ and $p+q=b$. Split middle term, factor by grouping.

■ **FAST METHOD:** $ac=-6$, $sum=5$: try 6 and -1 . Split: $2x^2+6x-x-3=2x(x+3)-(x+3)=(2x-1)(x+3)$.

Q14. Correct Answer: C [MEDIUM]

STEP-BY-STEP SOLUTION:

$x^3-4x=x(x^2-4)=x(x-2)(x+2)$. ✓

Fully factored using GCF then difference of squares.

WHY EACH OPTION IS / IS NOT CORRECT:

A) Partially factored — (x^2-4) can be factored further.

B) $(x-2)^2=x^2-4x+4\neq x^2-4$.

C) ✓ CORRECT — CORRECT. Completely factored.

D) $=x^3+2x^2-2x-4\neq x^3-4x$.

■ **TRAP ALERT:** Don't stop at $x(x^2-4)$. x^2-4 factors further as $(x-2)(x+2)$.

■ **KEY INSIGHT:** Always factor completely — check each factor for further factoring opportunities.

■ **FAST METHOD:** $x^3-4x=x(x^2-4)=x(x+2)(x-2)$.

Q15. Correct Answer: C [MEDIUM]

STEP-BY-STEP SOLUTION:

$$3x^2-12x=0 \rightarrow 3x(x-4)=0.$$

$$3x=0 \rightarrow x=0. \text{ Or } x-4=0 \rightarrow x=4. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) Missing the second solution $x=4$.

B) Missing $x=0$.

C) ✓ CORRECT — CORRECT. Two solutions from $3x(x-4)=0$.

D) Sign error: $(x-4)=0$ gives $x=+4$, not -4 .

■ **TRAP ALERT:** When factoring out x , $x=0$ is ALWAYS one solution. Don't discard it.

■ **KEY INSIGHT:** Factor out GCF before solving. Each factor set to 0 gives a solution.

■ **FAST METHOD:** $3x(x-4)=0 \rightarrow x=0$ or $x=4$.

Q16. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

$$6x^2-7x-3. \text{ AC method: } ac=6 \times (-3)=-18, p+q=-7.$$

$$\text{Find } p, q: pq=-18, p+q=-7 \rightarrow p=-9, q=2.$$

$$6x^2-9x+2x-3=3x(2x-3)+1(2x-3)=(3x+1)(2x-3). \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $(2x-3)(3x+1)=6x^2+2x-9x-3=6x^2-7x-3. \checkmark$

B) Same as A.

$$\text{C) } =6x^2-18x+x-3=6x^2-17x-3 \neq 6x^2-7x-3.$$

$$\text{D) } =6x^2-6x+3x-3=6x^2-3x-3 \neq 6x^2-7x-3.$$

■ **TRAP ALERT:** For trinomials with $a \neq 1$, always use the AC method or trial-and-error carefully.

■ **KEY INSIGHT:** $AC=-18$, $\text{sum}=-7$: try -9 and 2 . Group: $3x(2x-3)+1(2x-3)=(3x+1)(2x-3)$.

■ **FAST METHOD:** $pq=-18, p+q=-7$: $(-9)(2)$. Grouping $\rightarrow (3x+1)(2x-3)$.

Q17. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

$$x^2+3x-28=0. \text{ Find } p, q: pq=-28, p+q=3.$$

$$p=7, q=-4: 7 \times (-4) = -28, 7 + (-4) = 3. \checkmark$$

$$(x+7)(x-4)=0 \rightarrow x=-7 \text{ or } x=4. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $(x+7)(x-4)=0$.

B) Wrong signs: would be $(x-7)(x+4)$.

C) Product $4 \times 7 = 28 \neq -28$.

D) Product $(-4)(-7) = 28 > 0$, but we need product $= -28$.

■ **TRAP ALERT:** When $c < 0$, the two roots have OPPOSITE signs. Here $pq = -28$ means one positive, one negative root.

■ **KEY INSIGHT:** $pq = -28 \rightarrow$ opposite signs. $|p+q| = 3$ but $\text{sum} = +3 \rightarrow$ larger magnitude must be positive: $+7$.

■ **FAST METHOD:** Factors of -28 summing to 3 : 7 and -4 . Roots: $x = -7$ and $x = 4$.

Q18. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

$4x^2 - 12x + 9$: check if perfect square.

$$(2x-3)^2 = 4x^2 - 12x + 9. \checkmark$$

$$a=2x, b=3: (2x)^2 - 2(2x)(3) + 3^2 = 4x^2 - 12x + 9. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

$$\text{A)} = 4x^2 - 12x - 3x + 9 = 4x^2 - 15x + 9 \neq 4x^2 - 12x + 9.$$

B) ✓ CORRECT — CORRECT. Perfect square trinomial.

$$\text{C)} = 4x^2 - 2x - 18x + 9 = 4x^2 - 20x + 9 \neq 4x^2 - 12x + 9.$$

$$\text{D)} = 4x^2 - 4x - 9x + 9 = 4x^2 - 13x + 9 \neq 4x^2 - 12x + 9.$$

■ **TRAP ALERT:** Test: $(b/2a)^2 = (-12/8)^2 = (3/2)^2 = 9/4$. Multiply by a : $4 \times 9/4 = 9 = c$ ✓. It's a perfect square.

■ **KEY INSIGHT:** For $ax^2 + bx + c$ to be $(\sqrt{a}x \pm \sqrt{c})^2$, need $b^2 = 4ac$: $(-12)^2 = 144 = 4(4)(9) = 144$ ✓.

■ **FAST METHOD:** Is $(-12)^2 = 4(4)(9)$? $144 = 144$ ✓ $\rightarrow$ perfect square. $(2x-3)^2$.

Q19. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

$x^3 + 2x^2 - x - 2$: factor by grouping.

$$= x^2(x+2) - 1(x+2) = (x^2-1)(x+2).$$

$$= (x-1)(x+1)(x+2). \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. Fully factored.

B) $=x^3+0x^2-x-2 \neq x^3+2x^2-x-2$.

C) Partially factored: $x^2-1=(x-1)(x+1)$. Not fully factored.

D) $=x^3+x-2x^2-2 \neq x^3+2x^2-x-2$.

■ **TRAP ALERT:** After grouping, always check if any factor can be factored further (here $x^2-1=(x-1)(x+1)$).

■ **KEY INSIGHT:** Factor by grouping: split into pairs. Then check if the remaining quadratic factors.

■ **FAST METHOD:** $x^2(x+2)-1(x+2)=(x^2-1)(x+2)=(x-1)(x+1)(x+2)$.

Q20. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

$$(x-4)^2=x^2-8x+16.$$

Compare with $x^2-kx+16$: $-k=-8 \rightarrow k=8$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $k=4$: $(\text{half of } 4)^2=4 \neq 16$. Not perfect square.

B) ✓ CORRECT — CORRECT. $(x-4)^2=x^2-8x+16 \rightarrow k=8$.

C) $k=16$: $x^2-16x+16$. Not $(x-4)^2$.

D) $k=-8$ would give $x^2+8x+16=(x+4)^2$, not $(x-4)^2$.

■ **TRAP ALERT:** $(x-4)^2$ has middle term $-8x$, so the coefficient $k=8$ (positive).

■ **KEY INSIGHT:** Expand $(x-4)^2=x^2-8x+16$. Match coefficients: $k=8$.

■ **FAST METHOD:** $k=2 \times 4=8$.

Q21. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

$$2x^2-3x-2=0. \text{ Factor: } ac=-4, \text{ sum}=-3.$$

Find p, q : $pq=-4$, $p+q=-3 \rightarrow p=-4$, $q=1$.

$$2x^2-4x+x-2=2x(x-2)+1(x-2)=(2x+1)(x-2)=0.$$

$$2x+1=0 \rightarrow x=-1/2. \quad x-2=0 \rightarrow x=2. \quad \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $(2x+1)(x-2)=0$.

B) Signs wrong on both roots.

C) Both can't be roots: $2(4)-3(2)-2=0$ ✓ but $2(4)+6-2=12 \neq 0$.

D) $2(1/4)-3/2-2=0.5-1.5-2=-3 \neq 0$.

■ **TRAP ALERT:** For $2x+1=0$: $x=-1/2$ (not $x=1/2$). Solve each factor equation carefully.

■ **KEY INSIGHT:** Factor by grouping or AC method. Then set each factor to zero.

■ **FAST METHOD:** $(2x+1)(x-2)=0 \rightarrow x=-1/2$ or $x=2$.

Q22. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

$9x^2-6x+1$: check if perfect square.

$$(3x-1)^2=9x^2-6x+1. \checkmark$$

$$a=3x, b=1: 9x^2-6x+1 \checkmark.$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) $=9x^2+6x+1$. Wrong sign on middle term.

B) ✓ CORRECT — CORRECT. $(3x-1)^2=9x^2-6x+1. \checkmark$

C) $=9x^2-9x-x+1=9x^2-10x+1 \neq 9x^2-6x+1$.

D) $=9x^2-1$ — difference of squares, not the given expression.

■ **TRAP ALERT:** The negative middle term $(-6x)$ signals $(3x-1)^2$, not $(3x+1)^2$.

■ **KEY INSIGHT:** $(a-b)^2=a^2-2ab+b^2$. Negative middle $\rightarrow$ use minus: $(3x-1)^2$.

■ **FAST METHOD:** $b^2=4ac$: $(-6)^2=36=4(9)(1)=36 \checkmark \rightarrow$ perfect square. $(3x-1)^2$.

Q23. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

$$x(x-5)=14 \rightarrow \text{expand: } x^2-5x=14 \rightarrow x^2-5x-14=0.$$

$$\text{Factor: } pq=-14, p+q=-5 \rightarrow p=-7, q=2.$$

$$(x-7)(x+2)=0 \rightarrow x=7 \text{ or } x=-2. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $(x-7)(x+2)=0$.

B) Wrong signs.

C) Results from not expanding and rearranging.

D) $x=2$: $2(-3)=-6 \neq 14$.

■ **TRAP ALERT:** NEVER set $x=0$ or $x=5$ from $x(x-5)=14$. Move 14 to one side first. $x(x-5) \neq 14$ does not mean $x=14$ or $x-5=14$.

■ **KEY INSIGHT:** Set equation = 0 before factoring. $x(x-5)=14 \rightarrow x^2-5x-14=0$.

■ **FAST METHOD:** $x^2-5x-14=0 \rightarrow (x-7)(x+2)=0 \rightarrow x=7$ or $x=-2$.

Q24. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

$$12x^2+8x-15. \text{ AC method: } ac=12 \times (-15)=-180, p+q=8.$$

Find p,q: $pq=-180$, $p+q=8$. Try $p=18$, $q=-10$: $18 \times (-10) = -180$, $18 + (-10) = 8$. ✓

$12x^2+18x-10x-15=6x(2x+3)-5(2x+3)=(6x-5)(2x+3)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $(6x-5)(2x+3)=12x^2+18x-10x-15=12x^2+8x-15$. ✓

B) $=12x^2+20x-9x-15=12x^2+11x-15 \neq 12x^2+8x-15$.

C) $=12x^2-18x+10x-15=12x^2-8x-15 \neq 12x^2+8x-15$.

D) $=12x^2+9x-20x-15=12x^2-11x-15 \neq 12x^2+8x-15$.

■ **TRAP ALERT:** $AC=-180$: need factor pair summing to 8. Test 18 and -10.

■ **KEY INSIGHT:** AC method: $pq=-180$, $p+q=8 \rightarrow 18$ and -10 . Group and factor.

■ **FAST METHOD:** 18 and -10: $12x^2+18x-10x-15=6x(2x+3)-5(2x+3)=(6x-5)(2x+3)$.

Q25. Correct Answer: D [MEDIUM]

STEP-BY-STEP SOLUTION:

$x^4-81=(x^2)^2-9^2=(x^2-9)(x^2+9)$.

$x^2-9=(x-3)(x+3)$. x^2+9 doesn't factor over reals.

$x^4-81=(x-3)(x+3)(x^2+9)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $=(x^4-18x^2+81) \neq x^4-81$.

B) $=(x-3)^4 \neq x^4-81$.

C) Partially factored — (x^2-9) can be factored further.

D) ✓ CORRECT — CORRECT. Completely factored.

■ **TRAP ALERT:** (x^2-9) is NOT fully factored. It's a difference of squares: $(x-3)(x+3)$. Always factor completely.

■ **KEY INSIGHT:** x^4-81 : two-step factoring. Step 1: difference of squares. Step 2: factor x^2-9 again.

■ **FAST METHOD:** $x^4-81=(x^2-9)(x^2+9)=(x-3)(x+3)(x^2+9)$.

Q26. Correct Answer: C [HARD]

STEP-BY-STEP SOLUTION:

For $x^2+12x+c$ to be a perfect square: $c=(b/2)^2=(12/2)^2=6^2=36$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) 6 is $b/2$, not $(b/2)^2$.

B) Not $(12/2)^2=36$.

C) ✓ CORRECT — CORRECT. $(12/2)^2=36$. $x^2+12x+36=(x+6)^2$.

D) This is $b^2=144$, not $(b/2)^2=36$.

■ **TRAP ALERT:** The constant in a perfect square is $(b/2)^2$, NOT b^2 or $b/2$.

■ **KEY INSIGHT:** Completing the square: add $(b/2)^2$ to x^2+bx to make a perfect square trinomial.

■ **FAST METHOD:** $c=(12/2)^2=6^2=36$.

Q27. Correct Answer: A [HARD]

STEP-BY-STEP SOLUTION:

$$2x^3-16=2(x^3-8)=2(x-2)(x^2+2x+4). \checkmark$$

Using difference of cubes: $a^3-b^3=(a-b)(a^2+ab+b^2)$ with $a=x$, $b=2$.

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. Factor out 2, then difference of cubes.

B) Not fully factored.

C) $=2(x-2)(x^2+2x+4)$ ✓ but $(2x-4)$ is not fully simplified; should factor as $2(x-2)$.

D) $(x-2)^3=x^3-6x^2+12x-8 \neq x^3-8$.

■ **TRAP ALERT:** Factor out the GCF (2) first, THEN apply the difference of cubes identity.

■ **KEY INSIGHT:** Always factor out GCF before applying special factoring formulas.

■ **FAST METHOD:** $2(x^3-8)=2(x-2)(x^2+2x+4)$.

Q28. Correct Answer: B [HARD]

STEP-BY-STEP SOLUTION:

$$x^2-4x=21 \rightarrow x^2-4x-21=0.$$

Factor: $pq=-21$, $p+q=-4 \rightarrow p=-7$, $q=3$.

$$(x-7)(x+3)=0 \rightarrow x=7 \text{ or } x=-3. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) $x=3$: $9-12=-3 \neq 21$.

B) ✓ CORRECT — CORRECT. $(x-7)(x+3)=0$.

C) $x=-7$: $49+28=77 \neq 21$.

D) Product $(-7)(-3)=21 \neq -21$.

■ **TRAP ALERT:** Rearrange to $=0$ FIRST: $x^2-4x-21=0$. Then factor.

■ **KEY INSIGHT:** Always rearrange to standard form before factoring.

■ **FAST METHOD:** $x^2-4x-21=0 \rightarrow (x-7)(x+3)=0 \rightarrow x=7 \text{ or } x=-3$.

Q29. Correct Answer: B [HARD]

STEP-BY-STEP SOLUTION:

$$\text{Factor by grouping: } x^3+x^2-4x-4=x^2(x+1)-4(x+1)=(x^2-4)(x+1)=(x-2)(x+2)(x+1). \checkmark$$

Factors: $(x+1)$, $(x-2)$, $(x+2)$. Answer: $B=(x+1)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $p(1)=1+1-4-4=-6 \neq 0$. Not a factor.

B) ✓ CORRECT — CORRECT. $p(-1)=-1+1+4-4=0$ ✓.

C) $p(2)=8+4-8-4=0$ ✓. Also a factor! But B is listed first.

D) $p(-4)=-64+16+16-4=-36 \neq 0$.

■ **TRAP ALERT:** Both B and C are factors. The question asks 'which IS a factor' — $B=(x+1)$ is listed.

■ **KEY INSIGHT:** Grouping: $x^2(x+1)-4(x+1)=(x+1)(x^2-4)=(x+1)(x-2)(x+2)$.

■ **FAST METHOD:** Factor Theorem: $p(-1)=-1+1+4-4=0$ ✓ $\rightarrow (x+1)$ is a factor.

Q30. Correct Answer: A [HARD]

STEP-BY-STEP SOLUTION:

Roots $3/2$ and -4 with leading coefficient 2.

$$f(x)=2(x-3/2)(x+4)=(2x-3)(x+4)=2x^2+8x-3x-12=2x^2+5x-12=0. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $2x^2+5x-12=0$ has roots $3/2$ and -4 . ✓

B) Would have roots with product 6 and sum $5/2$.

C) Positive constant: product = $6 > 0$, so both roots same sign. But $3/2$ and -4 have opposite signs.

D) Roots: $3/2$ and -4 ? Sum = $3/2 - 4 = -5/2 \rightarrow -5x$ coefficient, but here $-5x$. Product = -6 . $2x^2-5x-12$: discriminant = $25+96=121 \rightarrow$ roots = $(5 \pm 11)/4 = 4$ or $-3/2$. Not our roots.

■ **TRAP ALERT:** Build from factors: $2(x-3/2)(x+4)$. Expand carefully: $(2x-3)(x+4)$.

■ **KEY INSIGHT:** From roots r_1, r_2 and leading coeff a : equation = $a(x-r_1)(x-r_2)$.

■ **FAST METHOD:** $2(x-3/2)(x+4)=(2x-3)(x+4)=2x^2+8x-3x-12=2x^2+5x-12=0$.

Q31. Correct Answer: D [HARD]

STEP-BY-STEP SOLUTION:

x^2-5x^2+4 : let $u=x^2$.

$$u^2-5u+4=(u-1)(u-4)=(x^2-1)(x^2-4).$$

$$=(x-1)(x+1)(x-2)(x+2). \checkmark$$

Both A and B are correct forms — A is partially factored, B is completely factored.

WHY EACH OPTION IS / IS NOT CORRECT:

A) Correct but not fully factored.

B) CORRECT and completely factored.

C) $=(x^2-8x^2+16) \neq x^2-5x^2+4$.

D) ✓ CORRECT — CORRECT. A is a valid factoring; B is the complete factoring. Both are correct.

■ **TRAP ALERT:** A is not wrong — it IS a valid factored form. B factors A further. 'Both' is the most complete answer.

■ **KEY INSIGHT:** A bi-quadratic substitution $u=x^2$ converts x^4-5x^2+4 into $u^2-5u+4=(u-1)(u-4)$.

■ **FAST METHOD:** $u=x^2$: $u^2-5u+4=(u-1)(u-4)=(x^2-1)(x^2-4)=(x-1)(x+1)(x-2)(x+2)$. Both A and B valid.

Q32. Correct Answer: A [HARD]

STEP-BY-STEP SOLUTION:

Double root $x=-3$ and simple root $x=1$.

$p(x)=(x+3)^2(x-1)=\dots$ expand.

$(x+3)^2=x^2+6x+9$.

$(x^2+6x+9)(x-1)=x^3-x^2+6x^2-6x+9x-9=x^3+5x^2+3x-9$.

Comparing with x^3+ax^2+bx+c : $a=5$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $p(x)=(x+3)^2(x-1)=x^3+5x^2+3x-9 \rightarrow a=5$.

B) Off — computed $(x+3)(x-1)^2$ perhaps.

C) This is the absolute value of the constant term.

D) Exceeds correct value.

■ **TRAP ALERT:** With a double root: use $(x-r)^2$ NOT $(x-r)$. So double root $-3 \rightarrow (x+3)^2$.

■ **KEY INSIGHT:** Double root at $r \rightarrow$ factor $(x-r)^2$ appears. Expand $(x+3)^2(x-1)$ carefully.

■ **FAST METHOD:** Sum of roots (Vieta's): $-3+-3+1=-5=-a \rightarrow a=5$.

Q33. Correct Answer: C [HARD]

STEP-BY-STEP SOLUTION:

$kx^2-8x+4=0$. For two distinct real solutions: discriminant >0 .

$\text{disc}=64-16k>0 \rightarrow 16k<64 \rightarrow k<4$.

Also $k \neq 0$ (otherwise not quadratic; becomes linear $-8x+4=0$ with one solution).

So $k<4$ and $k \neq 0$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Missing the restriction $k \neq 0$.

B) This gives $\text{disc}<0$ — no real solutions.

C) ✓ CORRECT — CORRECT. Need $\text{disc}>0$ AND quadratic ($k \neq 0$).

D) $k=2$ works ($\text{disc}=64-32=32>0$ ✓), but so does $k=-1$ ($\text{disc}=64+16=80>0$ ✓). Not $k>0$.

■ **TRAP ALERT:** Don't forget $k \neq 0$. If $k=0$, the equation becomes linear (not quadratic) with one solution.

■ **KEY INSIGHT:** For a quadratic ($k \neq 0$) with two distinct real solutions: $\text{disc} = b^2 - 4ac > 0 \rightarrow 64 - 16k > 0 \rightarrow k < 4$.

■ **FAST METHOD:** $\text{disc} = 64 - 16k > 0 \rightarrow k < 4$; AND $k \neq 0 \rightarrow$ answer C.

Q34. Correct Answer: A [HARD]

STEP-BY-STEP SOLUTION:

$$8x^3 + 27 = (2x)^3 + 3^3. \text{ Sum of cubes: } a^3 + b^3 = (a+b)(a^2 - ab + b^2). \\ = (2x+3)((2x)^2 - (2x)(3) + 3^2) = (2x+3)(4x^2 - 6x + 9). \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. Sum of cubes with $a=2x$, $b=3$.

B) Middle sign wrong: should be $-6x$ not $+6x$ in sum of cubes.

C) This is the difference of cubes $(2x)^3 - 3^3$.

D) $(2x+3)^3 = 8x^3 + 36x^2 + 54x + 27 \neq 8x^3 + 27$.

■ **TRAP ALERT:** Sum of cubes: $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$. The trinomial has a MINUS sign in the middle.

■ **KEY INSIGHT:** Sum: $(a+b)(a^2 - ab + b^2)$. Difference: $(a-b)(a^2 + ab + b^2)$. The signs alternate.

■ **FAST METHOD:** $8x^3 + 27 = (2x)^3 + 3^3 = (2x+3)(4x^2 - 6x + 9)$.

Q35. Correct Answer: A [HARD]

STEP-BY-STEP SOLUTION:

$x=1$ is a root $\rightarrow (x-1)$ is a factor. Divide $x^3 - 7x + 6$ by $(x-1)$.

Synthetic division: $[1, 0, -7, 6] \div (x-1)$.

$$\begin{array}{r|rrrr} 1 & 1 & 0 & -7 & 6 \end{array}$$

$$\begin{array}{r|rrrr} & 1 & 1 & -6 & \end{array}$$

$$\begin{array}{r|rrrr} & 1 & 1 & -6 & 0 \end{array}$$

Quotient: $x^2 + x - 6 = (x+3)(x-2)$.

Roots of $x^2 + x - 6 = 0$: $x = -3$ and $x = 2$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. Other roots from $x^2 + x - 6 = (x+3)(x-2)$.

B) $x=3$: $27 - 21 + 6 = 12 \neq 0$. Not a root.

C) $x=3$ is not a root (see B).

D) $x=3$ is not a root.

■ **TRAP ALERT:** Use synthetic division correctly with zero coefficient for x^2 (which is missing).

■ **KEY INSIGHT:** Missing x^2 term: include 0 as placeholder in synthetic division array.

■ **FAST METHOD:** Divide by $(x-1)$: quotient $x^2+x-6=(x+3)(x-2)$. Roots: -3 and 2.

TOPIC SUMMARY — FACTORING POLYNOMIALS

Digital SAT Math — Comprehensive Review

■ COMMON MISTAKES

- Difference of squares: $x^2-a^2 = (x+a)(x-a)$, NOT $(x-a)^2$. These are completely different!
- Sum/difference of cubes: the trinomial factor in $a^3-b^3=(a-b)(a^2+ab+b^2)$ has ALL POSITIVE signs. Students flip the middle sign.
- Never set $x(x-5)=14$ and conclude $x=14$ or $x-5=14$. ALWAYS rearrange to $=0$ first.

■ MUST REMEMBER

- Perfect square test: $(b/2)^2=c$. The constant in the perfect square is (half the middle coefficient) squared.
- AC method for ax^2+bx+c : find p,q where $pq=ac$ and $p+q=b$. Split the middle term and factor by grouping.
- Factor Theorem: $(x-r)$ is a factor $\leftrightarrow p(r)=0$. Use this to check factors quickly.
- Always factor out GCF BEFORE applying difference/sum of squares or cubes.

■ EXPECTED ON DSAT

- Factoring quadratics (simple and complex coefficients) is among the most tested topics — 3-4 questions per test.
- Solving by factoring (set each factor to zero) appears at every difficulty level.
- Difference of squares and perfect square trinomials are tested in both recognition and application forms.
- Sum/difference of cubes appears occasionally at Hard level.